

REPO**MIND** v3

Free local, minimal burst. The on-prem cost + trust layer that makes AMD the **cheaper enterprise choice** — and pulls Instinct + EPYC into the segment NVIDIA can't reach. Open-weight, air-gapped, 100% AMD.

-89%

TOKENS · MEASURED

-90%

CACHED INPUT · AMD-SERVED¹³

\$4.12

ACT I · 256K REPO · MI300X

\$1T

AMD AI-ACCEL. TAM BY 2030¹⁴

Small local model → delegates to Fireworks. Enterprise **cost-down**, on AMD.

You framed the north-star at kickoff: **a small local model that delegates to Fireworks for the heavy pipeline — free local compute, minimal paid burst.** REPOMIND is that pattern, built and measured.

MEASURED COST LAYER · MACBOOK + AMD-HOSTED FIREWORKS

AST skeleton — code exploration	−89%
Cached input — repeated repo prefix ¹³	−90%
Confidence cascade — skips the deep call	skip
Industry routing, no quality loss ¹²	40–85%

The app **streams the \$ saved per query** on screen — real, not projected.

THE DIFFERENTIATOR · FREE ON-PREM

Open-weight runs on the enterprise's **own AMD GPU → \$0 per-token**, data never leaves — ACT I proof: full **256K repo for \$4.12** of MI300X time. The paid Fireworks burst is then cut **−89/−90%**.

Free local + minimal remote = the enterprise TCO argument where **AMD wins**.

LOCAL TIER · IN PROGRESS

Gemma as the local router on the 48GB Radeon (wiring as GPU frees up) → the strongest Gemma-on-AMD story. Deep tier = open-weight on Fireworks (AMD Instinct).

Barred from cloud AI — or **bleeding** on it.

Path A • **Locked out** — \$0 AI today

ON-PREM IS THE ONLY LEGAL OPTION

Samsung — banned ChatGPT after code leak ¹	2023
JP Morgan — restricted ChatGPT firm-wide ²	2023
Apple — blocked ChatGPT + Copilot ³	2023
US Space Force — banned genAI / CUI ⁴	2023
EU AI Act — non-compliance penalty ⁶	≤€35M/7%

→ On-prem open-weight on their own AMD Instinct + EPYC.

Path B • **Token tax** — bleeding on cost

"THE SUBSIDY ERA ENDED"

Uber — 2026 AI-coding budget gone in 4 mo ⁷	\$1,500 cap
GitHub Copilot — usage billing \$29→\$750 ⁸	~25x
Cursor — \$2B ARR, neg. margin, built router ⁹	\$2B ARR
Model-API spend — doubled YoY ¹⁰	\$8.4B/yr
Enterprises — >\$250K/yr on LLM APIs ¹⁰	37%
Microsoft — cut internal Claude Code licenses over cost ¹⁸	Jun 2026

→ The router cuts 40–70% & migrates the cheap majority to AMD open-weight.

Every path pulls through AMD: on-prem = Instinct/EPYC; the router turns existing cloud-AI spenders into AMD-inference customers.

The \$1T-TAM workload for the segment NVIDIA's CUDA moat **can't defend.**

AMD'S OWN TAILWIND

Lisa Su sizes the AI-accelerator TAM at **\$1 trillion by 2030** (up from \$500B by 2028)¹⁴; Data Center **\$5.8B, +57% YoY** (Q1 FY2026)¹⁵, with **agentic AI** named as the driver. **REPOMIND is that agentic workload.**

"Software is tokens plus time, and speed is the moat." — Anush Elangovan, VP AI Software, AMD¹⁹

PULL-THROUGH — FROM A \$0 BASELINE

Each on-prem deployment ≈ **\$200–500K** Instinct + EPYC. REPOMIND is an **AI agent** (scheduling + tool-invocation) — the exact workload AMD says **drives CPU demand**¹⁷: EPYC just hit a record **46.2% server-CPU revenue share**, server-CPU TAM → **\$120–170B by 2030**, Lisa Su sees a 1:1 CPU:GPU ratio. This segment deploys **zero** AI today, NVIDIA **legally locked out** → adoption from a **\$0 baseline**. All on AMD's open stack (ROCm 7 + vLLM).

GEMMA-ON-AMD — THE REFERENCE GOOGLE LACKS¹⁶

Google positions Gemma 4 for "agentic workflows, coding assistants, regulated workloads on AMD ROCm" — but has **no named regulated on-prem production reference**. REPOMIND is it.

THE MARKET IT ROUTES — NESTED, SOURCED

Model-API spend now (measured) ¹⁰	\$8.4B/yr
Enterprise-LLM market by 2034 ¹¹	\$71B
Serviceable — both paths	\$0.4–4B
Routing cut, no quality loss ¹²	40–85%

Per-company & market \$ are modeled on the measured floor + cited public data, ranged, not booked. Fireworks is AMD-Instinct-powered (partner). Named companies are not customers.

The cost + trust layer for on-prem enterprise AI on AMD.

Proof, already shipped: AMD ACT I — 1st Place AI Agents + Outstanding Social Engagement (two trophies · 481 submissions · 10,770 builders) · verified on real MI300X **\$4.12 / 256K / single card** · **AMD Developer officially featured REPOMIND** · 3 Zenodo preprints. **lablab.ai:** *"one of the standout submissions — the business case is airtight."*

TIER 1 · CORE SHIPPED
On-prem agent · local→Fireworks cost routing ·
Ed25519 AI Flight Recorder · CUDA→ROCm · air-gap
· 38 tests + CI.

TIER 2 · TRUST NEAR-TERM
Cost/Savings FinOps · self-verifying answers · PII
gateway · continuous red-team · sovereign RAG.

TIER 3 · SOVEREIGN HORIZON
Sovereign AI-in-a-box · verifiable unlearning · AI-
decision insurance · trusted AI metering.

HONESTY IS THE MOAT
Every number measured or labeled modeled. In a
field of demos that die after the video, this one has a
real user and a real business case.

Sources — **1** Forbes '23 (Samsung) · **2** CNN '23 (JPM) · **3** TechCrunch '23 (Apple/Copilot) · **4** Air&Space Forces '23 (Space Force/CUI) · **6** artificialintelligenceact.eu (Art. 99) · **7** TechCrunch '26 (Uber cap) · **8** GitHub Blog/TechCrunch '26 (Copilot \$29→\$750) · **9** TechCrunch/Reuters '26 (Cursor) · **10** Menlo Ventures '25 (\$3.5B→\$8.4B API spend; 37%>\$250K) · **11** Global Market Insights (\$8.8B→\$71.1B) · **12** Orq/TrueFoundry/IBM '26 (routing 40–85%) · **13** Anthropic/OpenAI/Google docs (cache –90%) · **14** CNBC — Lisa Su, AMD Financial Analyst Day Nov '25 (\$1T by 2030) · **15** CNBC/DataCenterDynamics (AMD Q1 FY26 DC \$5.8B, +57%) · **16** blog.google — Gemma 4 · **17** AMD blog "Agentic AI Changes the CPU/GPU Equation"; Mercury Research / BofA — EPYC record 46.2% server-CPU revenue share, server-CPU TAM →\$120–170B by 2030, Lisa Su 1:1 CPU:GPU · **18** thenextweb / cryptobriefing '26 — Microsoft pulls internal Claude Code over cost · **19** Dev Interrupted / migovi '26 — Anush Elangovan (AMD VP AI Software): "software is tokens plus time, speed is the moat" · ROCm Everywhere · MI350P PCIe enterprise on-prem.